APP DEV-1 PROJECT REPORT
TREKKING MANAGEMENT

1. Student Details

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About Me: I am an dual degree student pursuing offline degree from IIT-Dh and online degree from IIT-M . I like to do research and create something useful for society . I enjoy creating frontend which are simple and easy to understand alongside being creative.

2. Project Details

Project Statement:
The Trekking Management Application is a web-based system developed to simplify the management of trekking activities. It allows users to explore and book treks, trek staff to manage assigned treks, and administrators to oversee the entire system. The application reduces manual effort by maintaining trek, booking, and user information digitally. It also provides secure role-based access for different types of users.

Approach:
The application is developed using the Flask framework following a modular and role-based architecture. SQLite is used as the database with SQLAlchemy ORM for data management, while HTML, CSS, Bootstrap, and Jinja2 are used to build the user

interface. Flask-Login provides secure authentication and session management. The project is organized into separate modules for authentication, admin, staff, and user functionalities, making it easy to maintain and extend.

3. AI/LLM DECLARATION

I used **ChatGPT** to assist in writing SQLAlchemy model definitions, debugging, few places in frontend for redesigning
The extent of AI/LLM usage is around **10–12%**, limited to **code suggestions and documentation formatting**.
All final implementation logic and integration were done manually.

4. TECHNOLOGY AND FRAMEWORKS USED

Technology / Library	Purpose
Flask	Core backend web framework used to build the web application and handle routing.
SQLAlchemy	Object Relational Mapper (ORM) used to interact with the SQLite database.
Jinja2	Template engine used for rendering dynamic HTML pages.
Bootstrap 5	Used for responsive UI design and styling of the application.

Technology / Library	Purpose
Flask-Login	Handles user authentication, session management, and role-based access.
HTML5, CSS3 & JavaScript	Used to create the frontend interface and basic client-side interactions.
SQLite	Lightweight local database used to store users, treks, bookings, and staff information.

5. DATABASE SCHEMA/ ER DIAGRAM

Tables

User – Stores user account details (id, name, email, password, role, approval status)

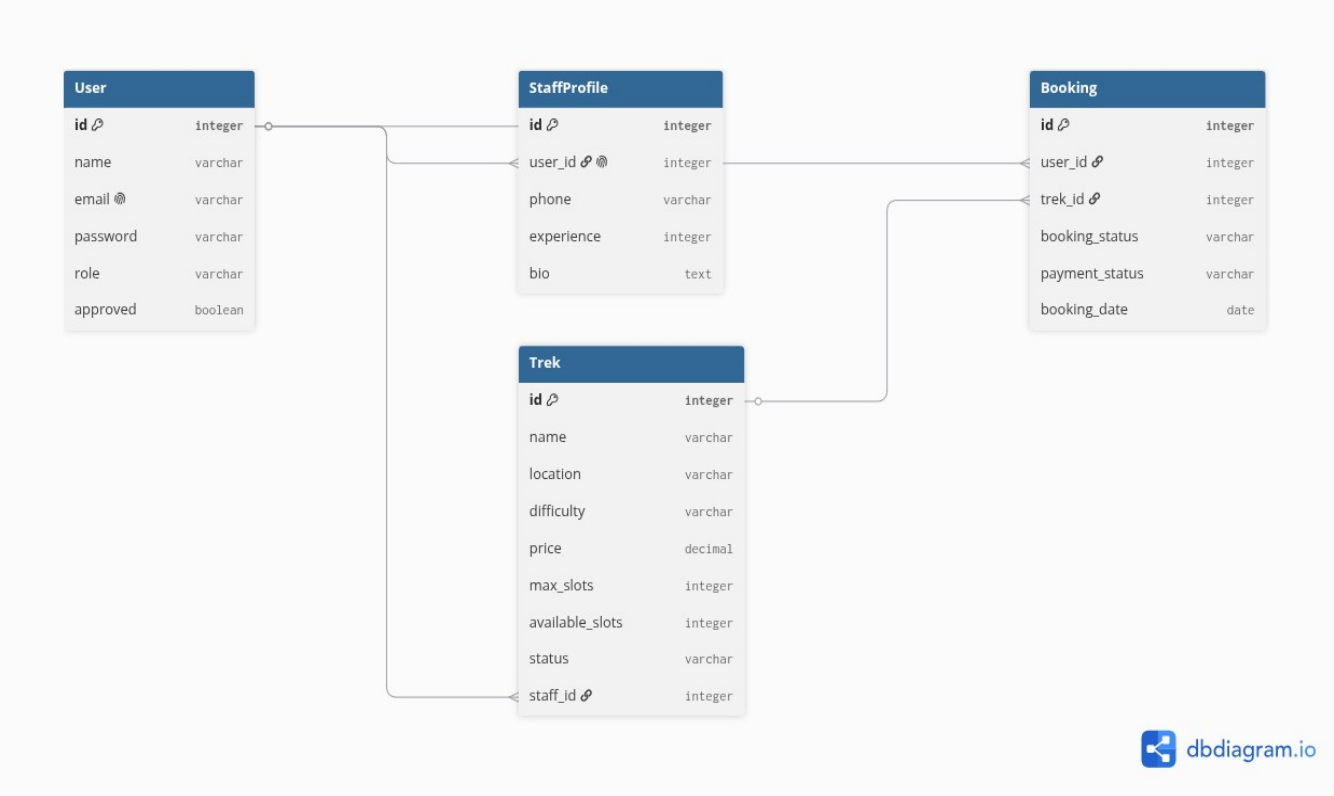
StaffProfile – Stores additional information for trek staff (id, user_id, phone, experience, bio)

Trek – Stores trek information (id, name, location, difficulty, price, max_slots, available_slots, status, staff_id)

Booking – Stores trek booking details (id, user_id, trek_id, booking_status, payment_status, booking_date)

Relationships

- **One-to-One → User ↔ StaffProfile**
- **One-to-Many → Staff (User) → Trek**
- **One-to-Many → User → Booking**
- **One-to-Many → Trek → Booking**



6. VIDEO PRESENTATION

https://drive.google.com/file/d/1kncHkz1J39bzW1HK9eLIXvtHd561mKN_/view